Statistics for Computer Scientists

Lecture 10: Regression and correlation I

21 November 2025

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Where are we

Lecture	Topic	Book
5 Sept	Introduction, statistics fundamentals	Chapter 1
12 Sept	Descriptive statistics	Chapter 2
19 Sept	Bivariate relationships	Chapter 3
26 Sept	Data collection	Chapter 4
10 Oct	Sampling distributions	Chapter 7
17 Oct	Estimation	Chapter 8
31 Oct	Significance testing	Chapter 9
7 Nov	Comparing two groups	Chapter 10
14 Nov	Categorical data analysis	Chapter 11
21 Nov	Linear regression	Chapter 12
28 Nov	Multiple regression	Chapter 13
5 Dec	Nonparametric statistics	Chapter 15
12 Dec	Assignment discussion	

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Today's lecture

- Recap lecture 9 (categorical variable associations)

- Regression models
- Prediction equation
- Residuals
- Correlation
- Inference (test of independence)

Recap categorical variable associations

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Categorical variable associations

- 2 groups and 2 outcome options: z-test for proportion
- More groups and/or outcome options: chi-squared test of independence
- Chi-squared: **is there** an association?
- Residuals: what is the **pattern** of association?
- Three options for **strength** of association:
  - Difference of proportions
  - Relative risk (= ratio of proportions)
  - Odds ratio

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Contingency Tables

sex		site 1	2	3	4	5	6	7	Total
f	Count	19.00	5.00	3.00	2.00	6.00	4.00	4.00	43.00
	Expected count	13.64	5.38	2.89	2.89	5.38	5.38	7.44	43.00
	Standardized residuals	2.29	−0.23	0.08	−0.71	0.38	−0.83	−1.81	
m	Count	14.00	8.00	4.00	5.00	7.00	9.00	14.00	61.00
	Expected count	19.36	7.63	4.11	4.11	7.63	7.63	10.56	61.00
	Standardized residuals	−2.29	0.23	−0.08	0.71	−0.38	0.83	1.81	
Total	Count	33.00	13.00	7.00	7.00	13.00	13.00	18.00	104.00
	Expected count	33.00	13.00	7.00	7.00	13.00	13.00	18.00	104.00

Chi-Squared Tests

	Value	df	p
X^2	7.545	6	.273
N	104		

Note. Continuity correction is available only for 2x2 tables.

Regression models

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Linear regression

- Use when both response and explanatory variables are **quantitative**
- X: a quantitative explanatory variable
- Y: a quantitative response variable

- Examples:
  - X is age, Y is height
  - X is number of years of education, Y is annual income

Linear regression

Questions to answer:

1. How can we **describe** the nature of the relationship between X and Y? (How can we predict Y from X?)
   - Regression equation and residuals
2. How **strong** is the association?
   - Correlation
3. Is the association statistically **significant**?
   - Test of independence

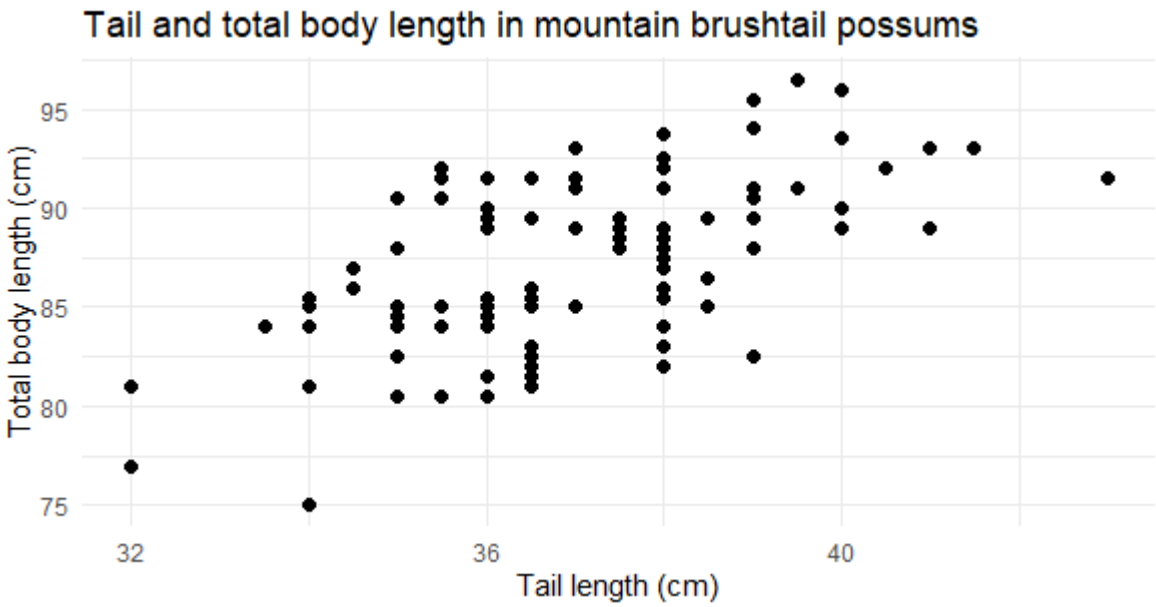
Linear relationships

- Linear function: $y = \alpha + \beta x$
- The y-value is expressed as a linear function of the x-value, with slope β and intercept α (two regression coefficients)

- Intercept: the predicted value of y when $x = 0$
- Slope: for each 1-unit increase in x, y increases by β units
  - $\beta = 0$: horizontal line, y does not depend on x
  - $\beta > 0$: line slopes upward, positive relationship
  - $\beta < 0$: line slopes downward, negative relationship

Linear relationship example

- Mountain brushtail possums
- X = tail length
  - Range: 32 – 43 cm
  - Mean: 37 cm
  - Standard deviation: 2 cm
- Y = total body length
  - Range: 75 – 90 cm
  - Mean: 87 cm
  - Standard deviation: 4 cm



Linear relationship example

- Relationship can be approximated by: $y = 42.6 + 1.2x$

- At $x = 0$, predicted body length:

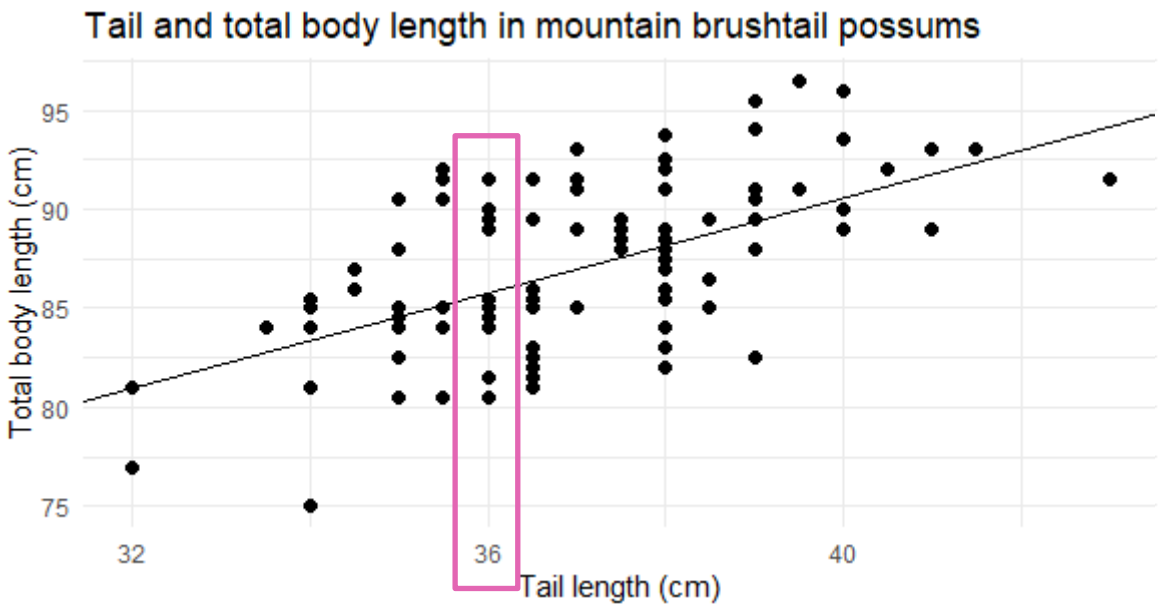
  $y = 42.6 + 1.2 * 0 = \mathbf{42.6}$ cm

  - Hypothetical!

- At $x = 36$, predicted body length:

  $y = 42.6 + 1.2 * 36 = \mathbf{85.8}$ cm

  - Actual y varies!



Effect of variable coding

- Slope and intercept depend on the units of measurements

- Example was in cm: $y = 42.6 + 1.2x$
- Transform both X and Y to inches (divide both by 2.54)
  - $y = 16.8 + 1.2x$
- Or transform Y to meters (divide Y by 100)
  - $y = 0.426 + 0.012x$

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Regression models

- In practice, Y is not completely determined by X
- $\alpha + \beta x$ represents the **mean** of y-values at specific x
- Population regression equation: $\mu_y = \alpha + \beta x$

- μ_y is the **population mean** of Y, which is the mean of the probability distribution of Y at a specific value of x
- Formula describes the **population** (Greek letters!)

Regression models and plausibility

- A **regression function** describes how the mean of the response variable Y changes when the value of explanatory variable X changes
- A **linear regression function** is part of a model for summarizing a relationship
  - A linear model is OK if the true relationship is approximately linear

- We use data to:

1. Check that the linear model is plausible (scatterplot)
2. Estimate the model parameters

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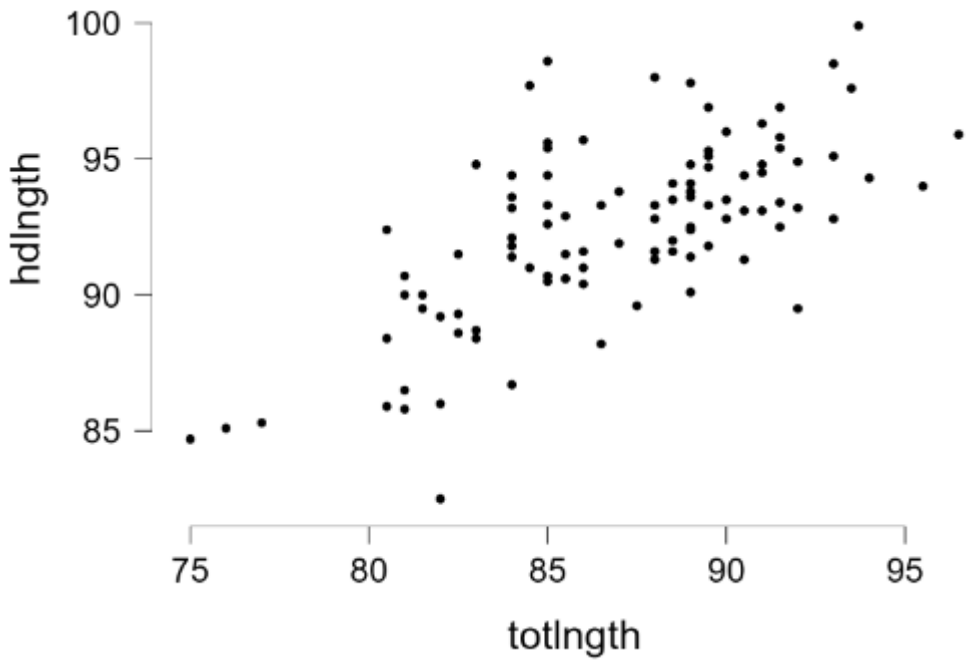
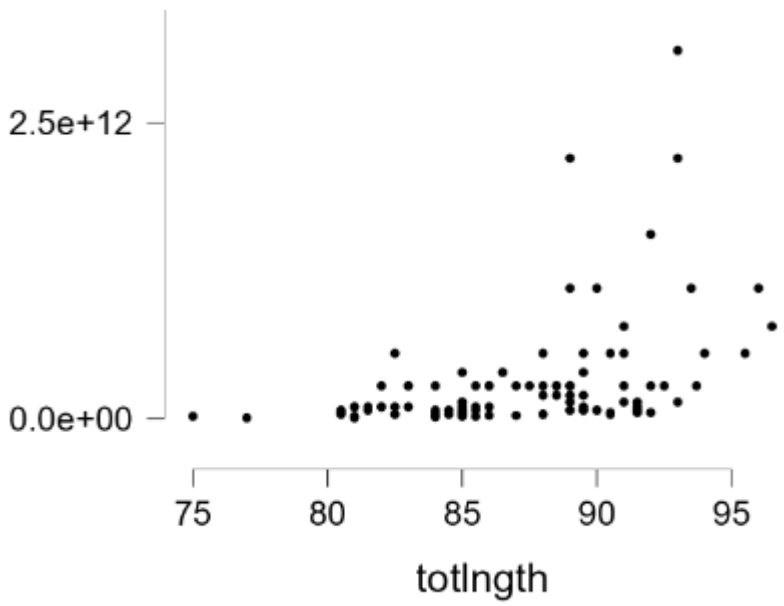
Plausibility

- Two quantitative variables: first make a **scatterplot**
- Visual analysis to see if a linear model seems plausible





Least squares prediction equation

- Population model: $\mu_y = \alpha + \beta x$
- Estimated by prediction equation: $\hat{y} = a + bx$
  - a estimates α, b estimates β, $\hat{y}$ estimates y

- a and b can be estimated from data

$$b = \frac{\sum(x_i - \bar{x})(y_i - \bar{y})}{\sum(x_i - \bar{x})^2}$$

$$a = \bar{y} - b\bar{x}$$

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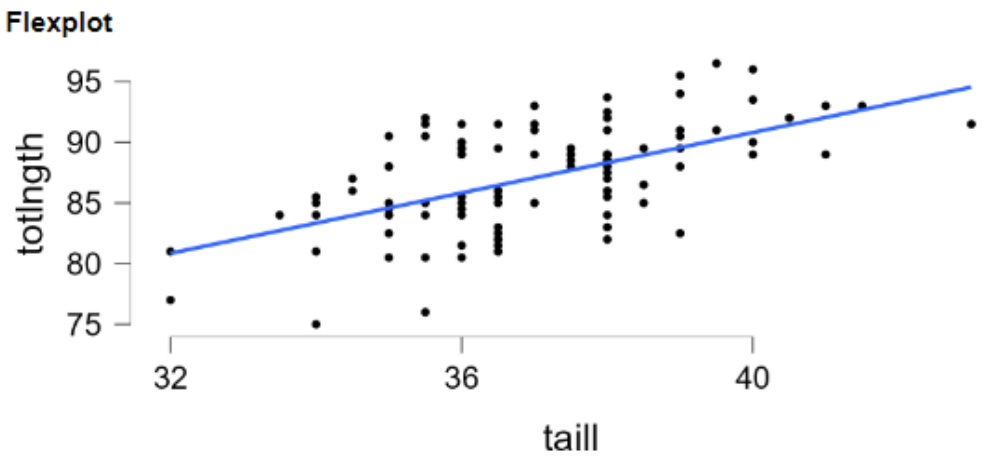
Example results + interpretations

Coefficients ▼

Model		Unstandardized	Standard Error	Standardized	t	p
M_0	(Intercept)	87.088	0.423		206.037	< .001
M_1	(Intercept)	41.037	6.657		6.165	< .001
	taill	1.244	0.180	0.566	6.927	< .001



- $a = 41$ and $b = 1.2$
- Regression formula: $\hat{y} = 41 + 1.2x$

- The predicted body length for a tail length of 0 is 41 cm
- 1-unit (cm) increase in tail length leads to increase of 1.2 in **predicted** body length
- $b > 0$, positive association between the variables **in this sample**

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Residuals

- Remember from chi-squared: $(f_o - f_e)$
- Difference between observed and expected

- Regression: residual is difference between observed y and predicted $\hat{y}$

- Example: subject 1 has $x = 40$, $y = 80$
- Prediction: $\hat{y} = 41 + 1.2 * 40 = 89$
- Residual: $y - \hat{y} = 80 - 89 = -9$

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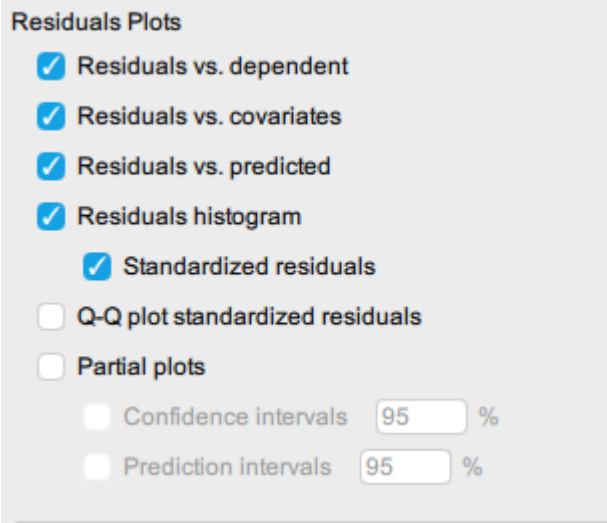
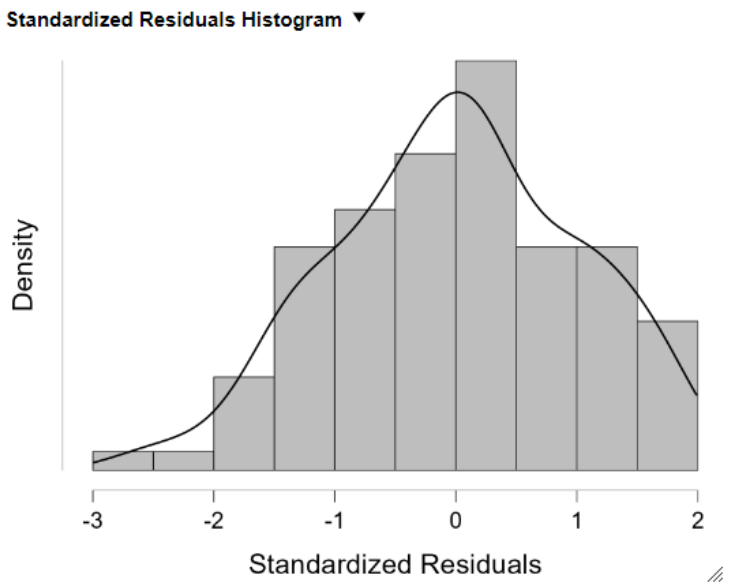
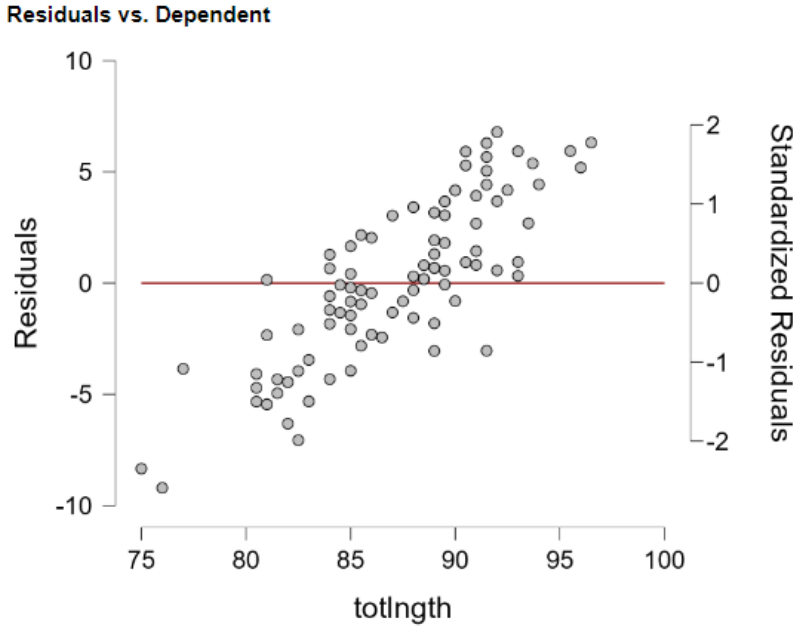
Residuals

- Mean of all residuals is always 0 (because of least-squares line!)
- Inspection of residuals important for regression inference







Residual standard deviation

- Linear regression model $\mu_y = \alpha + \beta x$ describes the **mean** of y at each x
- This is the mean of the conditional distribution of y given x
- The variability of this distribution is the conditional or **residual standard deviation**
  - "How much does y vary if I already know x?"

- Estimate of the residual standard deviation of y is:
- $s = \sqrt{\frac{residual\ SS}{n-2}} = \sqrt{\frac{\sum(y_i - \hat{y})^2}{n-2}}$
- $n - 2$ is the degrees of freedom!

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Residual standard deviation

- $Residual\ SS = \sum(y_i - \hat{y}_i)^2 = 1301.5$
- $n = 104$
- $s = \sqrt{\frac{residual\ SS}{n-2}} = \sqrt{\frac{\sum(y_i - \hat{y})^2}{n-2}} = \sqrt{\frac{1301.5}{102}}$
  $= \sqrt{12.8} = 3.6$

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
M_1	Regression	612.3	1	612.34	47.99	< .001
	Residual	1,301.5	102	12.76		
	Total	1,913.8	103			

Note. M_1 includes taill

Note. The intercept model is omitted, as no meaningful information can be shown.

Descriptives ▼

	N	Mean	SD	SE
totlngth	104	87.09	4.311	0.423
taill	104	37.01	1.960	0.192

- Interpretation: at any fixed level of tail length, the estimated standard deviation of body length is 3.6.

- Compare with (not conditional) standard deviation of y, which was 4.311!

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More formulas

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
M_1	Regression	914.063	1	914.063	93.257	< .001
	Residual	999.763	102	9.802		
	Total	1913.826	103			

Note. M_1 includes hdlngth

Note. The intercept model is omitted, as no meaningful information can be shown.

- Mean square error: $\text{MSE} = \frac{residual\ SS}{n-2}$
  - residual standard deviation: $s = \sqrt{MSE} = \sqrt{\frac{residual\ SS}{n-2}}$
- $df = n - 2$, because we estimated two parameters (α and β)

- Total sum of squares: $Total\ SS = \sum(y_i - \bar{y})^2$
- Consists of two things: **residual sum of squares** (SSE) and **regression sum of squares** (RSS)
- $Total\ SS = \sum(y_i - \hat{y}_i)^2 + \sum(\hat{y}_i - \bar{y})^2$

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Example - possums

- Mountain brushtail possums, $n = 104$
- X: tail length
- Y: total body length

- Software output:

RSS = regression sum of squares = $\sum(\hat{y}_i - \bar{y})^2$

SSE = sum of squared errors = residual sum of squares = $\sum(y_i - \hat{y}_i)^2$

TSS = total sum of squares = $\sum(y_i - \bar{y})^2$ (independent of regression!)

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
M_1	Regression	612.3	1	612.34	47.99	< .001
	Residual	1,301.5	102	12.76		
	Total	1,913.8	103			

Note. M_1 includes taill

Note. The intercept model is omitted, as no meaningful information can be shown.

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Example

- Prediction equation: $\hat{y} = 41 + 1.2x$
- Estimate of conditional standard deviation: $s = \sqrt{\frac{SSE}{n-2}} = \sqrt{\frac{1301.5}{102}} = 3.6$
- Predicted body lengths vary around a mean of $41 + 1.2x$ with a *sd* of 3.6 cm
- At tail length $x = 36$ cm, total body length is estimated to be distributed with

  $\bar{y} = 41 + 1.2 * 36 = 84.2$ cm and $s = 3.6$ cm.
- Marginal *sd* of total body length is larger!

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
M_1	Regression	612.3	1	612.34	47.99	< .001
	Residual	1,301.5	102	12.76		
	Total	1,913.8	103			

Note. M_1 includes taill

Note. The intercept model is omitted, as no meaningful information can be shown.

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Example

- Estimated conditional standard deviation (previous slide): 3.6
- Marginal standard deviation?

- $s_y = \sqrt{\frac{\sum(y_i - \bar{y})^2}{n-1}} = \sqrt{\frac{Total\ SS}{n-1}} = \sqrt{\frac{1913.8}{103}} = 4.3$

- Variation in body length among possums with the same tail length is **less** than total variation in body length

ANOVA

Model		Sum of Squares	df	Mean Square	F	p
M_1	Regression	612.3	1	612.34	47.99	< .001
	Residual	1,301.5	102	12.76		
	Total	1,913.8	103			

Note. M_1 includes taill

Note. The intercept model is omitted, as no meaningful information can be shown.

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Sums of squares

- When we predict Y without using X, best predictor is the sample mean of Y, and the prediction error is: $Total\ SS = \sum(y - \bar{y})^2$
- When we do use X to predict Y (with the prediction equation), the prediction error is $Residual\ SS = \sum(y - \hat{y})^2$
- The difference between these two is the amount of error that the regression function "solves" ($Regression\ SS = \sum(y - \bar{y})$)
- As residual SS decreases (and total SS stays the same), predictions get better

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r-squared

- The proportional reduction in error of using X to predict Y (via the prediction equation) instead of using the sample mean is:
- $r^2 = \frac{Total\ SS - Residual\ SS}{Total\ SS} = \frac{\sum(y-\bar{y})^2 - \sum(y-\hat{y})^2}{\sum(y-\bar{y})^2}$

- Sometimes called the **coefficient of determination**,
- but more commonly just r-squared

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Correlation

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Correlation

- Slope describes the direction of association between x and y, and depends on the units of the variables

- Correlation is a **standardized slope** and does not depend on units

$$r = b\left(\frac{s_x}{s_y}\right)$$

- where s_x and s_y are the sample standard deviations of x and y

- Correlation is a **measure of association**

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Properties of the correlation

- r is a standardized slope in sense that r reflects what b would equal if $s_x = s_y$

- $-1 \leq r \leq 1$, with r having same sign as b

- When all sample points fall exactly on prediction line, $|r| = 1$

- $r = 0$ when $b = 0$
  - can also happen when there is an association, but not linear

- The larger the absolute value, the stronger the association

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Correlation $r = 0$

Correlation $r = -0.3$

Correlation $r = 0.5$

Correlation $r = -0.7$

Correlation $r = 0.9$

Correlation $r = -0.99$

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Example

Coefficients ▼

Model		Unstandardized	Standard Error	Standardized	t	p
M_0	(Intercept)	87.088	0.423		206.037	< .001
M_1	(Intercept)	41.037	6.657		6.165	< .001
	taill	1.244	0.180	0.566	6.927	< .001

- Correlation: $r = b\left(\frac{s_x}{s_y}\right) = 1.244 * \frac{1.960}{4.311} = 0.57$
- (moderate positive association)
- Then $r^2 = 0.57^2 = 0.32$

Descriptives ▼

	N	Mean	SD	SE
totlngth	104	87.09	4.311	0.423
taill	104	37.01	1.960	0.192

ANOVA ▼

Model		Sum of Squares	df	Mean Square	F	p
M_1	Regression	612.3	1	612.34	47.99	< .001
	Residual	1,301.5	102	12.76		
	Total	1,913.8	103			

Note. M_1 includes taill

Note. The intercept model is omitted, as no meaningful information can be shown.

- $r^2 = \frac{1913.8 - 1301.5}{1913.8} = 0.32$
- 31% error reduction when using x to predict y, compared to using $\bar{y}$ to predict y!
- "31% of variation in total body length is explained by tail length"

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Properties of r^2

- Because $-1 \leq r \leq 1, \quad 0 \leq r^2 \leq 1$
- Minimum possible $Residual\ SS = 0$, in which case all sample points lie on the prediction line, and $r^2 = 1$
- r^2 does not depend on units, or which variable is X and which is Y

- If $b = 0$, then $\quad a = \bar{y} - b\bar{x} = \bar{y}$ (intercept is sample mean)
- So for any x, $\quad \hat{y} = a + bx = \bar{y}$ (prediction is sample mean)
- And so $Total\ SS = Residual\ SS$ and $r^2 = 0$ (regression doesn't add anything)

- Wooclap

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Inference

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Linear regression

Questions to answer:

1. How can we **describe** the nature of the relationship between X and Y? (How can we predict Y from X?)

   - Regression equation and residuals

2. How **strong** is the association?

   - Correlation

3. Is the association statistically **significant**?

   - Test of independence

Inference about slope β or correlation ρ

1. **Assumptions**

- Data were gathered using randomization
- There is a linear relationship between X and Y in the population
- The standard deviation σ of the conditional distribution of Y is the same at each X value
- The conditional distribution of Y is normal at each value of X *(not so important for two-sided test with large n)*

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Test of independence

2. **Hypotheses**

- Parameter: population slope in regression model β
- Estimator: least squares estimate b

- $H_0: \beta = 0$ (X and Y are independent)
- $H_a: \beta \neq 0$ (X and Y are dependent)
  - Or one-sided: $\beta > 0$ or $\beta < 0$

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Test of independence

3. **Test statistic**

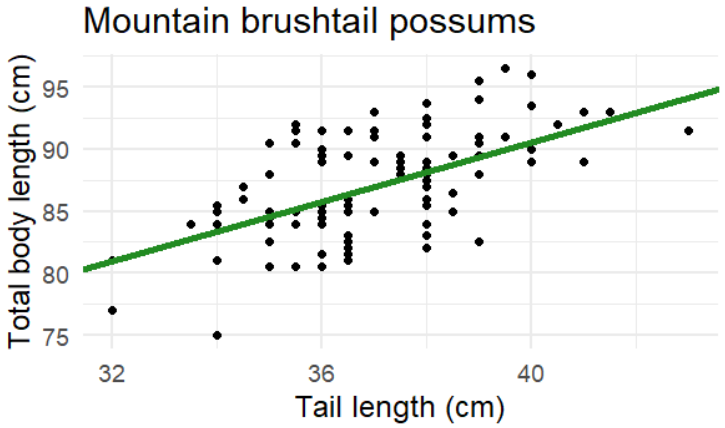
$t = \frac{b - 0}{se}$, $\quad se = \frac{s}{\sqrt{\sum(x-\bar{x})^2}} = \frac{s}{s_X\sqrt{n-1}}$

with $df = n - 2$

4. **P-value and/or conclusion**

As in t-test, use t-table and relate back to context!

Example - possums



- $t = \frac{b-0}{se} = \frac{1.244}{0.180} = 6.927$ (also given by software table)
- $df = n - 2 = 104 - 2 = 102$
- $p \ll 0.001$

- There is considerable evidence against $H_0: \beta = 0$
- There is a positive relationship between tail length and total body length in mountain brushtail possums

Coefficients

Model		Unstandardized	Standard Error	Standardized	t	p
M_0	(Intercept)	87.088	0.423		206.037	< .001
M_1	(Intercept)	41.037	6.657		6.165	< .001
	taill	1.244	0.180	0.566	6.927	< .001

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Confidence interval for slope

- Instead of significance test, CI for β can be calculated
- General formula: $b \pm t * se$
- t-score comes from Table B, use the column with $t_{0.5\alpha}$ and row with $df = n - 2$
- Calculation of se is the same as in the significance test: $se = \frac{s}{\sqrt{\sum(x-\bar{x})^2}}$

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Example - possums

- $b = 1.244, \quad se = 0.18$
- $df = 104 - 2 = 102$, we want a 95% CI so $t_{0.025} = 1.984$

- 95% CI for β is $b \pm t * se = 1.244 \pm 1.984 * 0.18 = 1.244 \pm 0.36 = (0.84, 1.56)$
- Because the entire interval is above zero, we conclude that the association in the population is positive, and the slope is between 0.84 and 1.56 (with 95% confidence)

Coefficients

Model		Unstandardized	Standard Error	Standardized	t	p
M_0	(Intercept)	87.088	0.423		206.037	< .001
M_1	(Intercept)	41.037	6.657		6.165	< .001
	taill	1.244	0.180	0.566	6.927	< .001

Test of independence

- Same test as for β can be done for ρ (population correlation)

- Test statistic: $t = \frac{r}{\sqrt{\frac{1-r^2}{n-2}}}$

- Same *t*-value, same *df*, same p-value

Extrapolation

- Linear regression is a model: we don't expect an **exact** linear relationship with constant variability, but in practice it is often a good approximation
- However, extrapolation beyond the observed range of x-values is dangerous!

- In the possum example, observed tail lengths were between 32 and 43 cm, all of adult possums. The model doesn't necessarily apply to a newborn possum with a 10 cm long tail.

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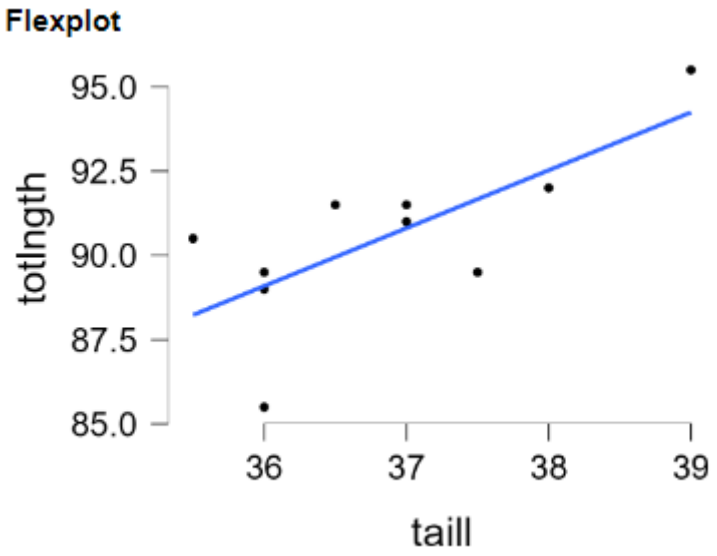
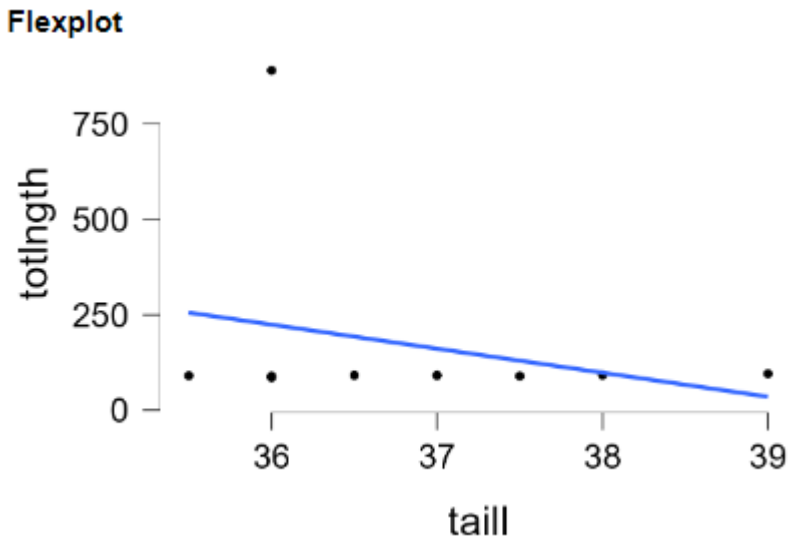
Outliers

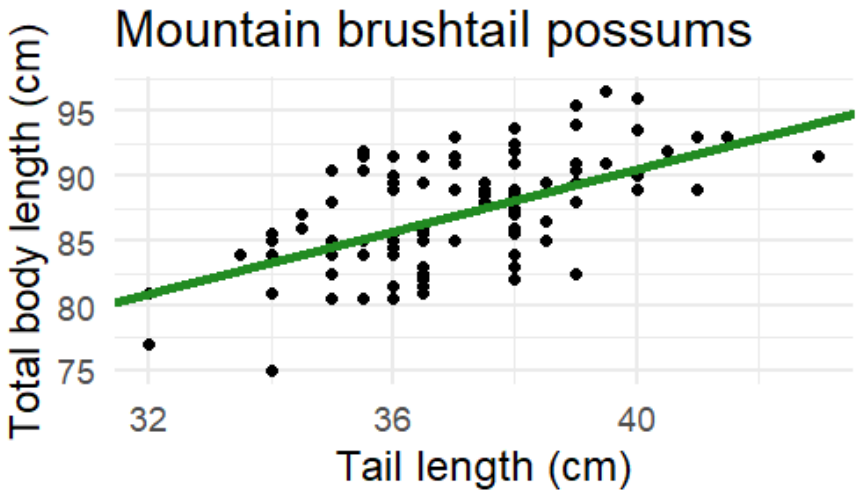
- Outliers are **influential observations**
- Their x and/or y-value are extreme and fall far from the linear trend seen in the rest of the data
- Also increases least squares!



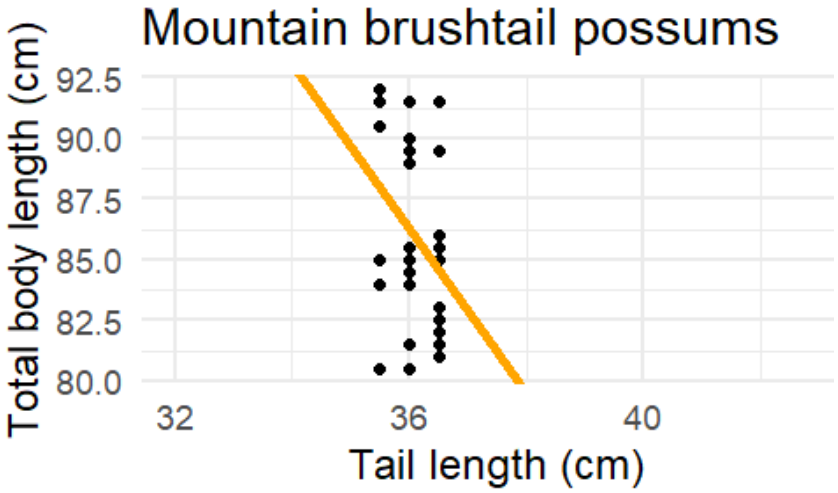


Effect of sampling range

- When the range of x-values in the sample is narrow, correlation is biased downward
- Therefore, correlation is mainly useful when both X and Y are randomly sampled from the population



range of x: 32 – 43 cm



range of x: 36 – 37 cm

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Wrapping up

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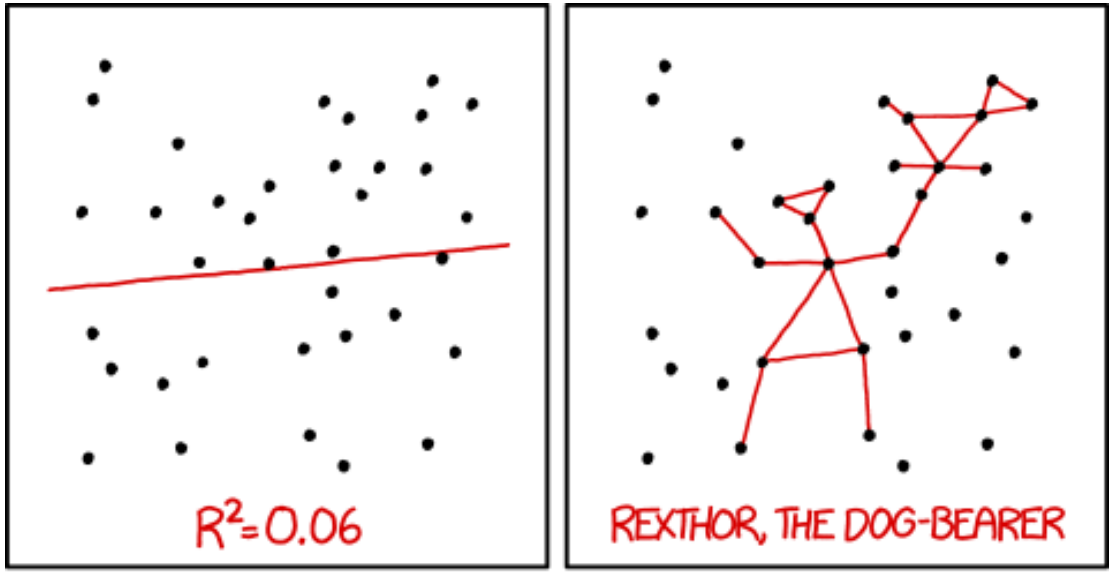
Wrapping up

- Read chapter 12 of the book

- JASP assignment 2 online
- Deadline: 7 December 23:59h

- Today: two quantitative variables
- Next week: more quantitative variables! Multiple regression



I DON'T TRUST LINEAR REGRESSIONS WHEN IT'S HARDER TO GUESS THE DIRECTION OF THE CORRELATION FROM THE SCATTER PLOT THAN TO FIND NEW CONSTELLATIONS ON IT.

https://xkcd.com/1725/